Do We Need Learnable Classifiers? A Hyperspectral Image Classification Algorithm Based on Attention-Enhanced ResBlock-in-ResBlock and ETF Classifier

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***Abstract*— Hyperspectral image (HSI) classification plays an important role in the field of remote sensing. Even though we can easily acquire hyperspectral remote sensing images, obtaining a large number of labeled hyperspectral samples remains challenging, especially in high-altitude or uninhabited areas. In this article, we propose a hyperspectral classification scheme for scenarios with insufficient labeled samples. This scheme is based on a variant of the ResBlock and a nonlearned classifier. First, we introduce a new and simplified backbone network for feature extraction. This network primarily consists of an attention-enhanced ResBlock-in-ResBlock module, which utilizes nested residual modules to enhance nonlinear expression and further optimizes the network using channel attention. Building upon this foundation, we address the challenge of achieving optimal classification with limited labeled training samples, a scenario described by the neural collapse theory. To address this, we introduce the equiangular tight frame (ETF) classifier and the dot-regression loss into hyperspectral classification. We conducted extensive comparative experiments using three HSI datasets. The experimental results demonstrate that our algorithm achieves superior classification accuracy, especially when the training sample size is small, outperforming other state-of-the-art algorithms. Furthermore, our algorithm maintains a low number of parameters and an overall complexity level.**

***Index Terms*— Channel attention, equiangular tight frame (ETF) classifier, hyperspectral classification, no-learned classifier, ResBlock-in-ResBlock.**

I. Introduction

HYPERSPECTRAL images provide us with a wealth of spectral information by capturing extensive and continuous band spectral data. This information spans not only the visible spectral range but also extends to other bands, including infrared and ultraviolet, contributing valuable insights beyond what is available in traditional three-channel remote sensing images. Consequently, hyperspectral images (HSIs) play a pivotal role across various domains of Earth observation. For this reason, HSI classification holds particular significance across numerous applications. This technology involves employing specific methods to assign pixel-level labels to each pixel within an HSI. This labeling process provides distribution information of ground objects for subsequent tasks.

However, HSI classification algorithms face unique challenges compared with conventional three-channel image classification tasks. These challenges include the curse of dimensionality, limited labeled samples, and spectral variability. With an abundance of spectral bands coupled with insufficient labeled samples, hyperspectral classification tasks are notably more demanding [1], [2].

To improve the performance of HSI classification, researchers have explored spectral selection and feature selection methodologies. These methods aim to reduce the dimensionality of data and enhance classification accuracy. Band selection algorithms typically identify and retain informative bands while discarding redundant ones [3], [4]. However, these algorithms overlook the unique spectral characteristics of different features, leading to suboptimal performance. In response, various feature selection algorithms have been proposed, such as principal component analysis (PCA) [5]. In addition, traditional algorithms based on support vector machines (SVM) [6] and random forest (RF) [7] heavily rely on expert knowledge and manual feature design, resulting in significant information loss and limited adaptability to diverse datasets, especially in scenarios with scarce training samples, leading to compromised classification results.

In recent years, deep learning has achieved a series of breakthroughs [8], [9], which have also significantly impacted tasks, such as endmember extraction [10], target detection [11], change detection [12], and anomaly detection [13] in remote sensing images. In HSI classification, the innate capacity of deep neural networks to automatically extract distinct and discriminative features offers a significant advantage [14], [15], [16], [17], [18]. Various architectures, such as stacked autoencoders (SAEs) [19], recurrent neural networks (RNNs) [20], graph convolution networks (GCNs), 3-D convolutional neural network (CNN) [21], [22], and transformers [23], [24], have demonstrated remarkable performance. Among these,

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CNNs have received considerable attention due to their powerful nonlinear representation capabilities and parameter-sharing mechanism [25], [26], [27], [28], [29].

While CNNs have exhibited remarkable performance in HSI classification, their uniform treatment of all spectral bands can occasionally lead to suboptimal results [28]. In response, a series of attention-based HSI classification algorithms have emerged [30], [31], [32], [33], [34], [35].

In situations where there is an ample supply of labeled samples and a balanced distribution, the aforementioned deep learning-based algorithms exhibit impressive performance. However, practical hyperspectral classification scenarios often present a significant challenge due to the scarcity of labeled samples [36]. For instance, consider the application of hyperspectral imaging to explore or monitor changes in remote and uninhabited areas, such as Mount Everest. Under these circumstances, despite the capability of airborne or spaceborne hyperspectral imaging devices to capture vast amounts of HSI data, the formidable expense tied to acquiring a substantial number of labeled hyperspectral classification samples presents a significant hurdle. Furthermore, the intricate nature of hyperspectral data often entails scenarios where identical categories of ground objects exhibit different spectra, while different ground objects may share the same spectrum. This phenomenon compounds the difficulty of achieving accurate HSI classification, particularly when confronting constraints posed by limited labeled samples.

To address the challenge of achieving high accuracy with limited labeled training samples, researchers have proposed various solutions, including transfer learning-based algorithms [37], [38], few-shot learning-based algorithms [39], [40], and self-supervised-based algorithms [41], [42]. However, most of these algorithms either require additional datasets for pretraining or rely on a large amount of unlabeled data for self-supervised learning, followed by fine-tuning on labeled samples [39], [40], [43], [44], [45]. However, in practical applications, obtaining additional data or a substantial amount of high-quality unlabeled data is also a challenge.

In this article, we propose a novel algorithm for HSI classification that does not require additional samples. Our approach combines an attention-enhanced ResBlock-in-ResBlock architecture with a nonlearned equiangular tight frame (ETF) classifier. In contrast to other methods that heavily rely on data augmentation and other complex techniques, our strategy aims to enhance the classification performance of HSIs under conditions of limited samples by using a simpler network paired with the ETF classifier.

Considering the limitations imposed by the scarcity of labeled hyperspectral training samples, we avoid employing overly complex networks that might lead to overfitting, thereby hindering the attainment of improved hyperspectral classification performance. On the other hand, overly simplistic networks may possess restricted feature extraction capabilities. In response, we introduce a ResBlock-in-ResBlock network structure, involving the embedding of one ResBlock within another—inspired by the concept of residual networks [8]. This augmentation enhances the network's nonlinear representation capabilities. Building upon this foundation, we emphasize the significance of spectral information due to the demanding nature of hyperspectral classification tasks. Moreover, in the input features of the ResBlock, the number of channels typically far exceeds the spatial dimension's size. Consequently, we leverage channel attention to further amplify the learned residual features. Based on this refined attention-enhanced ResBlock variant, we formulate a hyperspectral classification framework. In addition, the spectral differences between similar ground objects and the similarities in spectra among different types of ground objects also present challenges to classification. In such scenarios, traditional learnable neural network classifiers often struggle to reach the optimal convergence point, leading to subpar performance. To address this issue, we introduce an ETF classifier into HSI classification and combine it with a preassigned nonlearning classifier to produce the final classification result. Furthermore, in the loss function section, we employ a point regression loss function that aligns better with the ETF classifier.

We have validated the efficacy of our proposed approach through comprehensive experiments conducted on multiple HSI datasets. Our primary contributions are outlined as follows.

1) We present a novel HSI classification algorithm tailored for scenarios with limited training samples, comprising a simplified residual variant and an ETF classifier.
2) Within the residual module, we strategically embed a ResBlock within another ResBlock, enhancing the nonlinear representation capabilities. Furthermore, recognizing the paramount importance of spectral information in hyperspectral classification and the channel dimension's dominance over the spatial dimension in the input to the residual module, we integrate channel attention to further amplify the extracted residual features.
3) Addressing the scarcity of labeled training samples in HSIs and the considerable discrepancies among similar spectra, we address the limitations of achieving optimal convergence conditions in classification. As a solution, we employ a fixed, unlearnable ETF classifier. This classifier entails direct initialization of the most promising classification vectors, making optimization primarily focused on enhancing feature learning. Building upon the ETF classifier, we opt for a point regression loss function, a departure from the commonly used cross-entropy loss, tailored to harmonize with the ETF classifier's characteristics.
4) We execute a series of experiments on various HSI datasets, with the experimental outcomes attesting to the superior performance of our proposed algorithm. In addition, our ablation experiments on each module underscore the effectiveness of the individual components we introduced.

II. Related Works

A. HSI Classification With Limited Samples

HSI classification is one of the most important tasks of HSI processing. In this article, we focus on the limited

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sample-based HSI classification. According to the literature [36], small sample hyperspectral classification algorithms include those based on transfer learning-based algorithms [37], [38], few-shot learning-based algorithms [39], [43], [46], data augmentation-based algorithms [47], and autoencoder-based algorithms [48]. In [37], researchers preferentially trained a classification network on additional data and then transferred the network to the hyperspectral classification task with limited samples. On the other hand, in [38], the bottom feature extraction network of the classification network is pretrained on another data source and transferred, while the top-level classifier network is trained with limited labeled samples. Some few-shot hyperspectral classification algorithms adopt a strategy similar to transformer learning, where they are first pretrained on an additional dataset and then fine-tuned for few-shot learning on hyperspectral samples [39], [43]. Data augmentation techniques involve generating additional training samples using various methods, such as utilizing labeled data, to alleviate the issue of insufficient sample quantity. In the study mentioned in [47], researchers employed a generative adversarial network (GAN) to generate extra training data, thereby enhancing the accuracy of few-shot classification in hyperspectral imaging with limited samples. In addition, in [44], researchers introduced a deep contrastive learning network. The network is first constructed by forming contrastive groups and training through contrastive learning. Simultaneously, pseudolabels are generated for unlabeled samples, and samples with higher confidence pseudolabels are selected for further training in the network.

B. Neural Collapse and ETF Classifier

ETF classifiers are grounded in the concept of neural collapse, a phenomenon commonly observed in neural networks [49]. Neural collapse occurs when, under a balanced distribution of the training dataset and as the training error approaches 0, the final layers of features from the same class converge to a single vertex. This convergence results in all class vertices merging with their respective classifier vectors, forming a simplex structure known as an ETF. The ETF comprises K vectors within an $\mathbb{R}^d$ space (where $d \leq K-1$), exhibiting properties akin to a simple isometric compact frame. Specifically, all vectors share the same L_2 norms, and the inner product between distinct vectors is $[1/(k-1)]$, corresponding to the maximal angle between K isogons. This framework defines the optimal geometric configuration for a classifier, balancing the minimization of intraclass disparities and maximization of interclass disparities [50].

Given that neural collapse inherently describes the features of the last layer and the optimal geometry of the classifier, the ETF classifier predefines this optimal structure by configuring the classifier to align with the neural collapse setup. This predefined structure resembles a simple ETF, conventionally defined as follows [49], [51]:

$$\mathbf{E} = \sqrt{\frac{N}{N-1}} * \mathbf{U}\left(\mathbf{I_N} - \frac{1}{N}\vec{1_N}\vec{1_N}^{\top}\right). \tag{1}$$

Here, $\mathbf{E} = [\vec{e_1}, \ldots, \vec{e_N}] \in \mathbb{R}^{d\times N}$ introduces a rotation matrix while adhering to the constraint $\mathbf{U}^{\top}\mathbf{U} = \mathbf{I_N}$, where $\mathbf{I_N}$ represents the identity matrix (d represents the dimensionality of the output features from the backbone network, while N denotes the number of output categories). Moreover, $\vec{1_K}$ denotes a vector consisting of ones. All vectors within matrix $\mathbf{E}$ share an equivalent L_2 norm, and each vector maintains an equal distance from one another, as described by the equation

$$\vec{e_{n_1}}^{\top}\vec{e_{n_2}} = \begin{cases} -\dfrac{1}{K-1}, & n_1 \neq n_2 \\ 1, & n_1 = n_2 \end{cases} \quad \forall\, n_1, n_2 \in [1, N]. \tag{2}$$

III. Proposed Algorithm

A. Overall Framework

The overall classification network is depicted in Fig. 1. In contrast to other hyperspectral networks, our proposed network comprises only two components: a feature extractor based on the residual module and an ETF classifier. The complexity of networks often results in suboptimal performance, such as overfitting, particularly when labeled samples are limited. To address this, our feature extractor consists of only one residual module. To augment the nonlinear representation capacity of the residual module and enhance network depth, we introduce a variation of ResBlock that incorporates an additional ResBlock. Emphasizing the significance of spectral features in hyperspectral classification, we further refine the features extracted by the residual module using channel attention.

Our approach incorporates the ETF classifier as a distinctive element. This classifier is designed to predefine the optimal collapse point, aiming to maximize the distance between samples of different classes while minimizing differences among samples within the same class, thereby improving classification performance. Distinguishing our method from conventional classification algorithms, which typically utilize cross-entropy loss, we introduce the dot loss, as previously proposed in [50] and [51], into HSI classification.

In the subsequent subsection, we will provide a detailed introduction to each module.

B. Attention-Enhanced ResBlock-in-ResBlock

In the practical task of HSI classification, acquiring the entire HSI data has become relatively straightforward. However, challenges arise in certain scenarios, such as uninhabited or high-altitude areas, where obtaining an ample number of labeled hyperspectral training samples proves to be notably challenging. When training hyperspectral classification networks with only a limited number of labeled samples, the use of overly complex neural networks often leads to problems, such as overfitting, resulting in suboptimal classification outcomes. Therefore, in this article, we have chosen to focus on a more streamlined approach, with the core module of our proposed network comprising a single residual module.

However, it is clear that when only one residual module is included, the neural network is too shallow, and the nonlinear representation ability is insufficient. Based on this, we embed one ResBlock into another ResBlock to improve the nonlinear

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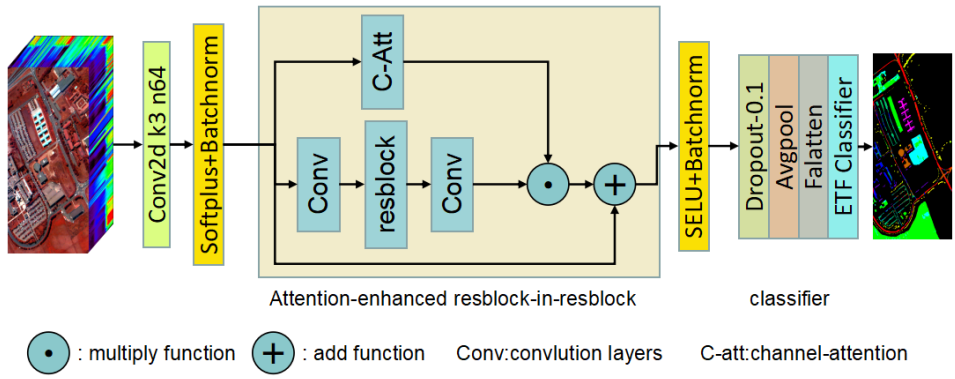


Fig. 1. Flowchart of our proposed framework is depicted below. As illustrated, our algorithm is exceptionally straightforward, featuring a backbone that solely incorporates a variant of the ResBlock. Meanwhile, the classifier utilizes an ETF classifier. In the backbone network, we utilize a ResBlock in ResBlock based on attention guidance to extract features from hyperspectral samples. The embedded residual module enhances the network's nonlinear expression capability, avoiding the issue of gradient vanishing. Simultaneously, spectral attention is employed to optimize the learned features. On the classifier side, the limited number of samples hinders the formation of an effective neural collapse structure. Therefore, we employ an ETF classifier, preassigning classification vectors and utilizing dot loss. This loss is designed to maximize the dissimilarity between different class samples while minimizing the differences within the same class, thereby enhancing classification results.

expression capability of the network. This process can be expressed as follows:

$$\mathbf{X} = \mathrm{Conv}_{1\times1}(\mathrm{Res}(\mathrm{Conv}_{1\times1}(\mathbf{X}))) + \mathbf{X} \tag{3}$$

where $\mathrm{Conv}_{1\times1}$ corresponds to a 2-D convolutional layer with a 1×1 kernel, augmented by the utilization of a LeakyReLU activation function. The symbol Res denotes a residual module encompassing a pair of convolutional layers, each equipped with a 3×3 kernel. This configuration can be mathematically defined as follows:

$$\mathrm{Res}(\mathbf{X}) = \mathrm{Conv}_{3\times3}(\mathrm{Conv}_{3\times3}(\mathbf{X})) + \mathbf{X}. \tag{4}$$

$\mathrm{Conv}_{3\times3}$ signifies a 2-D convolutional layer employing a 3×3 kernel, with a LeakyReLU activation function applied between successive layers in the hierarchy.

Nevertheless, given the paramount significance of spectral information in HSI classification tasks, it is common for the number of input feature channels within the residual module to considerably exceed the spatial dimension's size. In response, we employ channel attention to further amplify the acquired residual features. Specifically, we employ the efficient channel attention (ECA) [52] mechanism to facilitate this enhancement. The acquisition of attention can be mathematically formalized as follows:

$$\mathbf{A} = \delta(\mathrm{Conv1}d_3(G(\mathbf{X}))) \tag{5}$$

where $G(\cdot)$ signifies the global adaptive pooling algorithm and $\mathrm{Conv1}d_3$ denotes the utilization of 1-D convolution layers with a kernel size of 3. The symbol δ corresponds to the Sigmoid activation function. The outcome A represents the resulting attention weight, obtained through this process. Subsequently, the obtained weight A is harnessed to amplify the residual features gleaned from the compound residual module. This leads to a revised formulation of (3), presented as follows:

$$\mathbf{X} = \mathrm{Conv}_{1\times1}(\mathrm{Res}(\mathrm{Conv}_{1\times1}(\mathbf{X}))) \times \mathbf{A} + \mathbf{X}. \tag{6}$$

The flowchart of the channel-attention-enhanced ResBlock-in-ResBlock module is shown in Fig. 2.

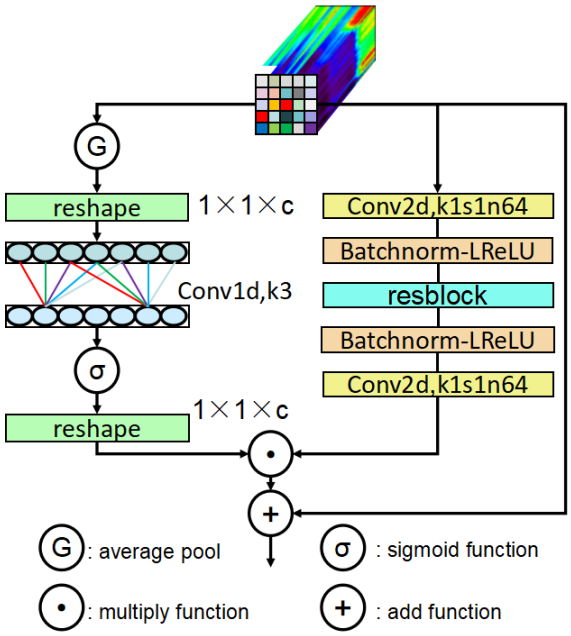


Fig. 2. Flowchart of the channel-attention-enhanced ResBlock-in-ResBlock module.

C. ETF Classifier and Dot-Loss Function

In hyperspectral classification tasks, challenges, such as insufficient labeled training samples, significant spectral variations within the same classes, and imbalanced sample distributions, can compromise the emergence of neural collapse. Consequently, achieving a well-defined neural collapse becomes challenging and may lead to a decline in performance. To address potential limitations arising from learning the last layer of the classifier under conditions of limited samples, hindering the establishment of an ETF structure, and ensuring the preservation of final HSI classification performance, we employ a nonlearning classifier, namely, the ETF classifier. Inspired by prior works [50], [51], we directly configure the classifier into an optimal ETF structure before

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network training. This approach helps avoid the issue of insufficient samples preventing the formation of a well-defined neural collapse phenomenon.

Aligning the classifier with the ETF definition as described in (1), we initiate the complete class space through a process of random initialization. During this initialization phase, where n' denotes the total number of categories, a foundational step is undertaken. Subsequent to this initialization, we proceed to construct an ETF classifier, denoted as $\hat{w}_{\text{ETF}}$, which encompasses class vectors for all possible class numbers.

The aforementioned ETF classification mechanism ensures the maximization of distances between classes in the final layer, while simultaneously minimizing distances within classes, thereby attaining an optimal classifier. In this context, following findings from the literature [51], when employing cross-entropy loss for classification, its gradient concerning the last layer of features comprises two distinct components. The first component is the "pull" term, wherein features attract the classification vector of the same class, while the second component represents the "push" mechanism, where features are propelled away from vectors of dissimilar categories. Notably, when the classifier vector is fixed to the optimal configuration, the "pull" term consistently aligns with the optimal direction, thus rendering the "push" term redundant. Building upon this insight, our classification framework adopts a specific loss known as the dot-regression loss. At this point, suppose the mapping learned earlier is

$$\mu = f(\mathbf{X}, \theta_f) \tag{7}$$

where f represents the preceding feature extraction network and μ represents the extracted feature information. For convenience of representation, we denote all modules in front of the ETF classifier as a whole as W_{ETF}. Then, the dot-regression loss can be expressed as follows:

$$L(\hat{\vec{\mu}}_i, W_{\text{ETF}}) = \frac{1}{2}(\hat{\vec{w}}_{y_i}^{\top}\hat{\vec{\mu}}_i - 1)^2 \tag{8}$$

where $\hat{\vec{\mu}}_i$ represents the normalized feature extracted from sample $\mathbf{X}_i$, i.e.,

$$\hat{\vec{\mu}}_i = \frac{\vec{\mu}_i}{\| \vec{\mu}_i \|} \tag{9}$$

where y_i is the class of sample $\mathbf{X}_i$ and $\vec{w_{y_i}}$ is the vector in the classifier W_{ETF} corresponding to class y_i. This is, the gradient of the loss function for the characteristic $\vec{\mu}_i$ is

$$\frac{\partial L}{\partial \hat{\vec{\mu}}_i} = -(1 - \cos\langle \hat{\vec{\mu}}_i, \hat{\vec{w}}_{y_i} \rangle)\hat{\vec{w}}_{y_i}. \tag{10}$$

According to the above formula, we can observe that the gradient consistently pulls the features in the direction of W, where W is the optimal vector preset by the ETF classifier itself and does not need to be updated. Ultimately, after the training process, the convergent spectral information will align with W, facilitating the attainment of an optimal geometric structure guided by the neural collapse phenomenon. This, in turn, leads to improved classification results.

TABLE I
CATEGORY INFORMATION OF THE IP DATASET

Class	Color	Class Name	Training	Validation	Test
1		Alfalfa	10	5	31
2		Corn-notill	10	5	1413
3		Corn-mintill	10	5	815
4		Corn	10	5	222
5		Grass-pasture	10	5	468
6		Grass-trees	10	5	715
7		Grass-pasture-mowed	10	5	13
8		Hay-windrowed	10	5	463
9		Oats	10	5	5
10		Soybean-notill	10	5	957
11		Soybean-mintill	10	5	2440
12		Soybean-clean	10	5	578
13		Wheat	10	5	190
14		Woods	10	5	1250
15		Building-Grass-Trees-Drives	10	5	371
16		Stone-Steel-Towers	10	5	78
		Total	160	80	10009

IV. EXPERIMENT AND ANALYSIS

To validate the effectiveness of the proposed algorithm, we perform experiments and analyses on several HSI datasets, namely, Indian Pines (IP), Kennedy Space Center (KSC), and WHU-Hi-HongHu (Honghu/HH) [53]. The IP and KSC datasets can be obtained from the website,[1] while the Honghu dataset is available on the website.[2] The IP dataset and KSC dataset were captured using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The IP dataset was collected over IP in June 1992. Researchers removed bands affected by noise and water absorption. The processed dataset used in the experiment comprises 200 bands with the dimensions of 145 × 145. This dataset encompasses 16 vegetation classes, with a total of 10 249 labeled pixels.

The KSC dataset was collected over the KSC in the United States in 1996. After eliminating certain noisy and water absorption bands, the KSC dataset comprises 176 bands for analysis. The spatial resolution of this dataset is 18 m per pixel, with the dimensions of 512 × 614 pixels. The dataset contains a total of 5211 labeled pixels, representing 13 classes of upland and wetland areas.

The Honghu dataset was captured using a 17-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro UAV platform on November 20, 2017, in Honghu, Hubei, China. This dataset represents a high-resolution hyperspectral remote sensing image dataset obtained through unmanned aerial remote sensing. It features a spatial resolution of approximately 0.043 m and has the dimensions of 940 × 475 pixels. The dataset encompasses 270 bands and includes 22 distinct feature categories.

The division of training, valid, and testing samples for the three datasets is illustrated in Tables I–III.

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2] http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE II
CATEGORY INFORMATION OF THE KSC DATASET

Class	Color	Class Name	Training	Validation	Test
1		Scrub	10	5	746
2		Willow-swamp	10	5	228
3		Cp-hammock	10	5	241
4		Cp/O-hammock	10	5	237
5		Slash-pine	10	5	146
6		Oak/B-Bammock	10	5	214
7		HW-swanp	10	5	90
8		Graminoid-marsh	10	5	416
9		Spartine-marsh	10	5	505
10		Cattail-Marsh	10	5	389
11		Salt-marsh	10	5	404
12		Mud-flats	10	5	488
13		Water	10	5	912
		Total	130	65	5006

TABLE III
CATEGORY INFORMATION OF THE WHU-HI-HONGHU DATASET

Class	Color	Class Name	Training	Validation	Test
1		Red roof	10	5	14026
2		Road	10	5	3497
3		Bare soil	10	5	21806
4		Cotton	10	5	163243
5		Cotton firewood	10	5	6203
6		Rape	10	5	44542
7		Chinese cabbage	10	5	24088
8		Pakchoi	10	5	4039
9		Cabbage	10	5	10804
10		Tuber mustard	10	5	12379
11		Brassica parachinensis	10	5	11000
12		Brassica chinensis	10	5	8939
13		Small Brassica chinensis	10	5	22492
14		Lactuca sativa	10	5	7341
15		Celtuce	10	5	987
16		Film covered lettuce	10	5	7247
17		Romaine lettuce	10	5	2995
18		Carrot	10	5	3202
19		White radish	10	5	8697
20		Garlic sprout	10	5	3471
21		Broad bean	10	5	1313
22		Tree	10	5	4025
		Total	220	110	384119

A. Evaluation Metrics

To assess the performance of different algorithms in HSI classification and comprehend the influence of various parameters, we conducted a thorough evaluation that encompassed both quantitative and qualitative analyses. In terms of quantitative assessment, we employed a variety of metrics, including overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and class-specific accuracy.

OA represents the ratio of accurately classified samples to the total sample count. AA is the average accuracy calculated across all land cover categories. The kappa coefficient provides a comprehensive measure of agreement between classification outcomes and the actual categories. Class-specific accuracy assesses the precision of each specific category, which is particularly crucial for datasets characterized by imbalances.

In addition to quantitative assessment, we conducted a subjective visual evaluation. We visually represented the classification outcomes of each algorithm and compared them with the ground-truth labels. This approach allowed for a visual appraisal of the performance and effectiveness of the different algorithms.

To guarantee a fair comparison, we also investigated additional factors, including training/testing time, the number of model parameters, and FLOPs. These aspects offer insights into the computational efficiency and complexity of the different classification algorithms.

B. Experimental Setting

All experiments were performed on a workstation equipped with an Intel[3] Xeon[3] Silver 4210 CPU clocked at 2.20 GHz and a single NVIDIA GeForce RTX 2080 GPU. The system environment used was Ubuntu 9.3.0-10ubuntu2. The proposed algorithm was implemented in Python 3.6 using the PyTorch 1.10 framework. For training, the Adam optimizer was employed with a learning rate of 0.001. A batch size of 32 was used for training across all datasets.

Given the limitations of limited labeled samples in training, to ensure a fair comparison, all algorithms underwent ten random repetitions. In each iteration, the same seed was used to ensure consistent training and test sets. The final measurement was obtained by calculating the average outcome and variance across these repetitions. In the tabulated results, the most outstanding outcomes are indicated in bold. For patch-based algorithms, a spatial window size of 11×11 was consistently utilized for 3-D patches. Training encompassed 150 epochs with a learning rate of 0.01, and a batch size of 32 was maintained.

C. Comparative Experiments

To evaluate the effectiveness of the proposed algorithm, we conducted comprehensive experiments, comparing it against 12 benchmark methods. These include two classical machine learning-based algorithms and ten state-of-the-art deep learning algorithms. The classical machine learning approaches chosen for comparison are RF [54] and SVM with a radial basis function (RBF) kernel [55]. The deep learning-based model includes CntextNet [26],[4] SSAN [30], RSSAN [31],[5] SSAtt [32],[6] A2S2K [22],[7] SSTN [34],[8] SSSAN [33], SSFTT [35],[9] CVSSN [56],[10] and AMS [29].[11] The chosen comparative algorithms were selected based on the studies conducted by Li et al. [29], [56], which offer an extensive overview of prevalent techniques in HSI classification. These encompass classical machine learning approaches,

[3]Registered trademark.
[4]https://github.com/eecn/Hyperspectral-Classification
[5]https://github.com/liereerniu/RSSAN-Hyperspectral-Image
[6]https://github.com/weecology/DeepTreeAttention
[7]https://github.com/suvojit-0X55aa/A2S2K-ResNet
[8]https://github.com/zilongzhong/SSTN
[9]https://github.com/zgr6010/HSI_SSFTT/
[10]https://github.com/lms-07/CVSSN
[11]https://github.com/lms-07/AMS-M2ESL

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TABLE IV

IP DATASET CLASSIFICATION RESULTS. EACH CATEGORY UTILIZES TEN SAMPLES FOR TRAINING, FIVE FOR VALIDATION DURING THE TRAINING PROCESS, AND THE REMAINING SAMPLES SERVE AS TEST SAMPLES. THE BEST RESULTS FOR OA, AA, AND KAPPA ARE DISPLAYED IN BOLD, WHILE THE SUBOPTIMAL RESULT IS UNDERLINED

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	SSFTT	CVSSN	AMS	Proposed
1	29.01±5.82	18.56±3.53	38.29±22.18	45.45±23.04	31.63±9.93	80.67±10.20	47.59±20.19	73.12±15.22	65.67±11.99	81.40±13.98	71.17±16.24	**85.88±13.29**	73.02±12.19
2	42.67±9.52	43.55±2.73	45.51±13.87	45.89±10.56	44.07±8.27	66.84±6.35	72.06±7.54	78.23±8.80	58.88±11.23	70.73±6.03	73.18±7.15	71.56±5.75	**82.72±4.84**
3	37.56±13.11	33.71±8.46	38.77±11.45	45.99±13.41	33.21±4.24	67.76±10.27	60.54±14.82	66.36±11.55	57.60±6.76	71.18±10.78	71.86±11.27	72.20±11.16	**81.15±11.67**
4	19.65±4.77	24.60±3.10	26.77±18.70	45.59±17.38	20.86±7.19	56.80±9.05	57.40±13.21	67.77±9.75	58.86±8.82	58.38±16.19	64.40±9.88	62.96±16.49	**73.98±10.26**
5	63.78±8.17	62.30±7.74	63.08±14.92	54.48±11.41	51.03±7.69	83.28±15.09	92.27±4.04	93.42±3.03	85.95±16.70	87.51±8.67	92.06±4.52	**95.34±3.88**	95.08±4.60
6	82.36±4.14	87.20±3.73	78.28±19.81	73.44±6.18	67.20±7.66	94.26±4.90	95.82±2.87	95.16±2.61	93.54±4.06	91.99±2.82	**96.00±1.59**	92.87±2.45	93.87±3.03
7	13.49±9.05	22.76±11.87	19.26±13.55	39.59±17.70	18.38±4.88	21.45±11.71	40.11±27.52	23.41±9.01	30.33±16.96	40.30±15.79	37.56±12.67	39.79±17.99	**41.43±18.30**
8	90.78±5.47	96.48±1.29	92.58±15.11	96.76±3.23	90.47±5.75	99.06±0.74	93.48±8.72	97.52±1.72	98.73±1.28	98.22±2.48	98.26±2.02	98.24±0.93	**99.04±1.67**
9	2.89±1.03	8.41±3.90	8.22±8.66	11.81±7.53	4.83±1.52	9.43±7.50	10.58±5.50	8.77±9.19	15.59±10.91	16.14±5.81	19.83±13.91	**23.47±17.49**	19.42±14.05
10	42.98±0.84	49.44±6.21	44.32±13.44	48.51±9.70	38.68±3.84	57.52±8.80	56.92±10.62	66.50±8.63	51.24±4.03	61.95±6.03	66.58±10.40	59.44±9.75	**76.74±10.58**
11	63.12±4.90	65.89±4.84	65.05±3.88	73.84±10.11	65.48±4.82	77.64±6.40	78.16±8.29	89.89±5.91	74.33±6.80	78.97±4.09	83.25±6.79	84.32±4.99	**92.23±3.74**
12	26.75±6.43	32.32±3.28	34.65±7.97	36.87±7.61	31.06±6.85	48.51±7.16	52.46±14.19	66.46±12.86	44.75±12.58	66.59±10.31	65.06±5.75	70.33±8.81	**79.38±12.28**
13	76.68±4.31	83.43±3.46	59.04±19.36	66.62±19.50	57.25±10.11	84.23±9.35	84.11±7.22	85.11±7.31	**95.11±6.04**	83.80±7.42	88.85±10.23	89.85±5.01	83.36±7.59
14	91.09±3.81	90.20±4.56	80.57±9.16	88.43±3.52	85.28±3.19	95.56±2.28	94.80±4.60	97.27±1.15	93.69±6.48	96.53±2.49	**97.69±1.32**	95.44±2.45	95.89±1.95
15	38.34±6.67	42.41±7.98	51.98±16.08	69.09±9.89	46.82±12.89	70.02±7.69	69.00±15.92	76.09±13.56	70.25±9.89	74.12±11.62	74.50±8.89	77.23±11.75	**77.71±10.70**
16	64.70±18.64	87.63±10.64	69.09±13.15	56.71±11.64	52.67±23.78	70.92±8.75	48.40±25.97	65.59±11.78	**79.15±6.29**	68.76±9.20	70.03±12.16	67.54±4.72	61.77±12.43
OA	52.54±2.62	55.42±1.86	51.27±6.92	57.87±4.33	52.19±1.99	72.78±2.70	70.59±7.59	79.66±1.52	69.62±2.88	76.61±1.35	79.11±0.86	78.44±2.52	**85.40±1.57**
AA	49.12±2.10	53.06±1.35	50.97±4.70	56.19±5.30	46.18±3.19	67.75±1.78	65.86±5.07	71.92±2.45	67.10±2.14	71.66±2.05	73.14±1.60	74.15±1.85	**76.68±2.68**
Kappa	46.81±2.74	50.07±2.01	45.74±6.94	52.88±4.79	46.40±2.01	69.28±2.87	66.93±8.46	77.09±1.69	65.64±3.32	73.55±1.41	76.39±0.98	75.68±2.73	**83.46±1.75**

TABLE V

KSC DATASET CLASSIFICATION RESULTS. EACH CATEGORY UTILIZES TEN SAMPLES FOR TRAINING, FIVE FOR VALIDATION DURING THE TRAINING PROCESS, AND THE REMAINING SAMPLES SERVE AS TEST SAMPLES. THE BEST RESULTS FOR OA, AA, AND KAPPA ARE DISPLAYED IN BOLD, WHILE THE SUBOPTIMAL RESULT IS UNDERLINED

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	SSFTT	CVSSN	AMS*	Proposed
1	93.73±3.81	74.52±2.51	96.49±4.03	95.45±1.68	94.85±3.78	98.14±1.99	97.32±2.20	98.48±3.51	96.31±3.60	2.73±32.66	**99.38±0.83**	73.78±17.57	99.36±0.63
2	69.69±4.48	84.26±6.39	65.06±8.02	74.75±1.86	62.30±22.20	88.54±7.55	76.35±22.64	95.67±4.03	79.86±9.88	47.87±15.64	95.50±6.03	23.28±19.28	**98.29±1.97**
3	83.14±2.61	66.74±6.93	69.04±7.05	59.65±8.78	57.12±8.32	77.46±6.83	74.74±31.16	86.22±20.09	64.47±13.41	19.95±24.69	**90.87±8.35**	17.55±19.57	90.30±10.66
4	55.81±6.19	44.72±5.93	60.87±8.21	58.01±3.24	49.59±14.71	72.97±10.93	64.72±20.14	75.04±27.41	58.74±12.55	56.94±23.73	84.65±8.16	24.42±23.77	**86.10±15.11**
5	54.71±8.69	47.42±7.71	8.99±7.96	47.20±1.68	45.15±8.89	67.82±10.78	69.36±25.52	67.92±20.51	45.26±15.72	47.68±27.47	69.50±15.70	17.79±12.53	**73.65±18.77**
6	43.88±5.74	45.87±8.98	80.57±7.34	56.94±1.86	58.45±14.00	82.13±14.42	**91.97±8.50**	68.03±28.95	69.69±14.78	41.94±39.50	86.38±11.85	29.15±23.07	82.84±14.59
7	60.33±6.7	68.35±4.44	68.17±16.52	53.12±8.78	47.61±14.54	71.41±12.78	**77.15±20.26**	69.85±18.07	62.99±11.25	19.75±29.86	70.62±13.50	21.36±18.06	69.34±20.70
8	51.90±5.73	61.23±5.50	66.54±11.0	65.07±3.24	52.80±16.76	90.12±4.14	82.62±11.70	84.04±7.99	58.26±14.51	34.31±15.51	77.37±19.22	58.34±13.18	**91.05±10.12**
9	75.51±5.05	75.00±3.73	86.68±7.76	74.45±1.68	65.78±9.84	87.51±9.22	91.16±11.07	**96.51±4.73**	74.47±8.83	33.18±31.68	90.22±8.99	72.64±18.16	94.69±8.06
10	73.10±9.74	90.05±9.01	67.55±11.56	2.99±1.86	69.85±10.22	94.67±5.33	92.45±12.05	98.12±3.41	67.76±11.25	68.73±14.07	85.41±30.26	38.74±22.45	**100.00±0.00**
11	96.39±3.74	92.42±2.97	95.02±7.55	97.79±8.78	87.99±7.19	95.90±3.43	94.34±7.60	96.61±5.18	94.41±6.30	76.06±31.32	97.17±7.23	91.87±7.19	**97.83±3.65**
12	78.21±5.65	84.54±7.94	87.81±6.87	67.84±3.24	67.08±12.68	84.99±11.54	92.81±8.17	98.09±3.26	73.40±20.85	55.19±10.59	73.46±20.03	49.37±24.28	**96.82±4.14**
13	96.48±1.64	99.65±0.92	96.24±2.72	83.99±1.68	82.24±8.24	94.06±9.57	**99.66±0.51**	99.01±.59	92.65±5.28	85.21±7.40	82.52±31.69	72.60±14.95	99.58±0.20
OA	77.78±1.59	79.65±1.50	82.14±4.03	74.30±18.54	70.06±4.89	88.45±4.29	87.31±5.36	90.60±2.99	75.01±5.86	43.73±10.01	84.38±12.68	54.22±7.56	**93.44±1.86**
AA	71.76±1.60	73.44±1.74	76.85±2.75	70.55±16.81	64.68±4.29	85.06±2.83	84.98±6.45	87.20±4.79	72.18±2.67	47.35±9.21	84.85±8.69	45.23±7.05	**90.76±2.25**
Kappa	75.25±1.75	77.34±1.66	80.07±3.74	71.38±20.56	66.50±5.53	87.10±4.83	85.86±5.96	89.52±3.33	72.19±6.40	37.53±10.95	82.70±13.94	48.87±8.39	**92.68±2.07**

2-D CNN-based methods, 3-D CNN-based techniques, residual learning methods, and diverse attention-based algorithms (including transformer-based approaches).

1) Quantitative Accuracy Analysis: The quantitative results are presented in Tables IV–VI. In these tables, we compare the classification performance of our algorithm with other methods across three datasets. Notably, certain traditional machine learning techniques, such as RF and SVM, occasionally surpass the classification accuracy of some deep learning approaches when dealing with limited sample sizes. This phenomenon may stem from the susceptibility of certain deep learning models to overfitting in scenarios with extremely constrained sample sizes, resulting in subpar performance.

Furthermore, aside from our proposed algorithm, the SSTN algorithm consistently demonstrates commendable classification accuracy across all three datasets. It is evident that our algorithm outperforms other methods on all three datasets, exhibiting an improvement in OA of 5.74, 2.84, and 3.23 percentage points, respectively, when compared with SSTN, the second most accurate algorithm across the datasets.

At the same time, it should be noted that, across the three datasets, there is a significant variation in the classification accuracy among different classes. Some classes achieve accuracy close to 100%, while others, especially those with fewer samples, exhibit very low accuracy, even less than 50%. This is, in fact, an indication of class imbalance. When randomly selecting samples, with a lower total number of samples, there is a higher probability of randomly choosing ten samples that include poorly representative instances, leading to a more significant accuracy drop due to misclassification of individual samples during classification. However, from the experimental results, it is evident that our proposed algorithm achieves optimal classification performance in the majority of classes.

In the IP and Honghu datasets, both CVSSN and AMS algorithms exhibit relatively high classification accuracy, showcasing improvement over other methods except for

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TABLE VI

HONGHU DATASET CLASSIFICATION RESULTS. EACH CATEGORY UTILIZES TEN SAMPLES FOR TRAINING, FIVE FOR VALIDATION DURING THE TRAINING PROCESS, AND THE REMAINING SAMPLES SERVE AS TEST SAMPLES. THE BEST RESULTS FOR OA, AA, AND KAPPA ARE DISPLAYED IN BOLD, WHILE THE SUBOPTIMAL RESULT IS UNDERLINED

Class	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	SSFTT	CVSSN	AMS	Proposed
1	83.14±7.85	93.86±3.85	87.44±5.97	95.67±2.76	87.62±7.30	96.17±2.36	86.80±18.61	95.96±4.62	93.02±6.96	92.97±7.01	96.09±3.51	96.83±1.97	**98.88±1.11**
2	45.08±7.51	50.34±7.32	48.22±9.76	62.55±9.20	49.05±8.53	69.12±6.41	54.93±8.08	66.91±12.67	61.12±8.51	55.24±9.43	**71.27±10.78**	54.01±7.17	70.14±11.79
3	82.04±4.79	85.33±6.87	85.46±5.24	89.29±5.4	79.18±8.99	93.95±2.10	84.85±12.95	94.24±6.27	84.84±10.29	89.58±3.29	93.19±3.02	91.48±1.75	**96.01±4.31**
4	96.63±0.57	96.62±0.60	98.04±1.00	99.23±0.55	97.51±1.34	99.24±0.56	98.27±1.14	99.36±0.40	98.68±0.59	98.23±0.44	98.93±0.64	99.04±0.47	**99.36±0.41**
5	11.82±.97	13.88±2.75	11.69±5.62	20.33±8.34	11.83±5.37	18.50±7.33	20.74±6.67	43.86±24.22	20.11±13.92	18.70±5.58	31.87±17.23	30.54±7.00	**45.04±16.81**
6	80.93±4.19	86.73±3.91	80.87±3.49	93.14±2.82	85.72±4.00	92.68±2.19	89.28±7.20	**96.70±1.95**	91.47±3.43	89.96±3.31	94.18±2.64	94.50±1.90	95.92±2.16
7	59.45±14.50	68.51±9.62	60.95±11.15	75.63±11.95	65.33±13.79	84.66±6.75	71.63±16.06	90.75±3.72	74.84±12.35	76.93±7.52	86.78±5.42	84.36±7.74	**90.45±4.04**
8	6.22±1.56	7.79±2.67	12.31±2.78	16.75±3.04	11.74±2.23	22.19±4.36	27.03±11.42	32.26±8.22	13.32±2.84	35.53±10.21	32.26±8.34	**41.91±9.01**	40.98±12.76
9	80.96±7.77	92.89±4.84	86.66±6.72	95.41±4.15	85.39±7.76	97.40±2.18	81.77±22.54	**98.89±0.81**	93.71±6.00	85.78±10.47	96.68±1.71	95.63±0.93	98.10±1.01
10	27.78±10.00	37.33±11.42	46.86±7.13	58.62±8.07	49.62±9.32	73.43±6.96	73.83±15.31	82.79±10.68	56.96±12.18	66.53±8.23	82.39±4.96	84.40±6.15	**92.38±4.70**
11	17.19±5.49	22.60±8.49	31.02±6.21	36.98±10.40	27.88±6.73	53.94±11.07	31.34±7.36	64.05±11.78	40.33±10.60	63.96±11.47	62.82±14.19	**82.10±12.23**	76.50±11.58
12	36.12±8.32	36.01±8.85	42.60±6.01	57.40±7.51	41.56±5.96	60.78±8.24	51.63±14.81	67.23±10.39	37.82±11.83	45.53±4.75	64.76±10.77	68.19±7.60	**71.67±13.45**
13	51.96±6.94	54.06±6.90	67.85±11.05	71.99±8.95	65.48±7.91	75.81±5.69	73.25±12.07	73.41±8.15	60.89±7.15	69.66±7.24	78.60±7.12	80.57±6.26	**84.69±7.31**
14	44.71±19.29	57.14±18.12	36.08±9.51	65.85±10.96	41.78±12.53	73.65±7.22	68.41±12.08	85.13±10.65	65.72±11.71	60.89±10.18	76.37±8.62	72.44±10.72	**91.12±6.35**
15	3.00±1.21	6.48±4.30	6.57±2.66	54.40±26.75	9.88±5.10	47.82±12.90	8.09±3.51	67.49±22.44	29.45±18.85	31.48±9.80	67.12±11.55	52.15±9.97	**78.29±19.55**
16	75.82±7.02	83.65±3.53	73.67±7.60	88.04±6.28	71.69±7.95	89.59±3.64	86.98±11.63	97.28±4.05	88.98±5.05	92.55±3.32	95.09±1.66	**97.14±2.43**	96.66±3.13
17	37.67±3.13	43.52±5.99	32.22±6.53	48.36±12.94	34.81±4.37	50.32±8.82	59.36±13.83	70.28±11.48	53.02±16.05	70.99±9.91	76.81±11.64	85.22±6.41	**88.03±9.27**
18	13.90±2.51	18.26±3.45	26.46±6.32	36.01±11.37	20.59±7.66	43.61±7.75	52.05±18.05	58.51±16.08	27.79±9.77	45.53±9.92	45.27±13.77	59.70±10.26	**64.97±10.98**
19	42.67±9.31	59.98±8.04	68.82±9.92	75.27±9.60	51.49±16.38	79.36±7.15	76.17±9.23	**85.63±5.44**	74.67±11.43	69.54±7.68	80.37±8.03	82.09±11.26	85.25±5.95
20	11.40±3.51	25.82±6.27	47.43±13.26	69.87±17.19	42.07±9.89	64.56±8.45	39.36±17.35	77.24±5.27	55.28±11.96	45.48±8.00	65.60±11.62	50.68±10.76	**77.83±4.79**
21	8.87±10.70	8.31±1.15	8.25±2.37	13.04±4.41	7.38±2.13	12.79±3.04	19.11±8.57	22.33±7.66	10.64±3.02	23.30±5.57	20.09±6.95	**51.77±9.03**	30.05±8.92
22	19.71±3.91	19.64±4.61	20.08±7.11	33.13±9.39	22.51±8.83	33.90±9.32	23.56±8.61	**62.97±17.33**	34.22±10.21	39.21±9.61	43.89±17.10	35.10±11.96	53.93±16.53
OA	51.46±3.89	59.80±3.29	62.28±6.32	73.33±7.77	61.17±8.42	76.99±4.47	67.35±8.08	85.05±3.78	69.08±8.62	75.90±3.49	81.92±4.01	83.33±2.51	**88.28±2.74**
AA	42.59±1.79	48.58±1.24	49.30±1.94	61.68±3.50	48.19±3.64	65.16±1.81	58.11±4.34	74.24±1.98	57.58±4.81	62.16±2.26	70.92±2.65	72.27±2.31	**78.47±3.07**
Kappa	44.61±3.42	52.99±3.05	55.77±6.12	68.17±7.77	54.58±8.14	72.26±4.92	61.66±8.58	81.57±4.29	63.33±8.72	70.84±3.85	77.86±4.56	79.78±2.83	**85.49±3.24**

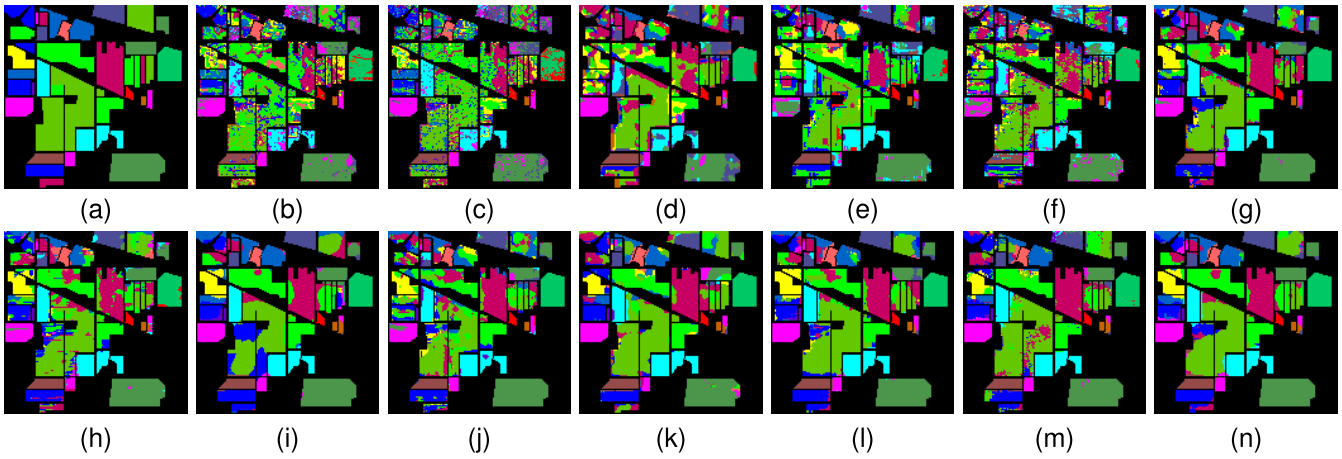


Fig. 3. Classification maps were generated for the IP dataset, utilizing only ten labeled training samples per class. The corresponding representations are as follows. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) SSFTT. (l) CVSSN. (m) AMS. (n) Proposed.

SSTN. However, during the course of our experimentation, we encountered a notable challenge with the AMS algorithm in the context of the KSC dataset. Remarkably, the AMS algorithm struggled to converge on the KSC dataset, yielding a classification accuracy of less than 10% even after attempting to reset the learning rate to 0.0001. These efforts only resulted in very poor classification performance.

We attribute this phenomenon primarily to the fact that the KSC dataset poses a unique challenge due to its scattered and relatively sparse sample distribution compared with the background. This peculiarity may have caused the AMS algorithm to learn features that are inadvertently biased toward regions without samples during the manifold learning phase, ultimately resulting in suboptimal performance.

We present the classification maps generated by all algorithms across the three datasets, as depicted in Figs. 3–5. In Fig. 3, a noticeable trend emerges: the classification maps produced by RF and SVM exhibit substantial instances of misclassification throughout the results. Conversely, the majority of misclassifications in the outcomes of deep learning algorithms are concentrated around the edges. Remarkably, our proposed algorithm stands out, boasting smoother edges and minimal instances of misclassification in the central region.

Transitioning to the KSC dataset, the presence of intersecting class areas is relatively scarce. Consequently, misalignment can result in the improper division of entire blocks. Among all algorithms, our approach notably demonstrates fewer misclassifications in this context.

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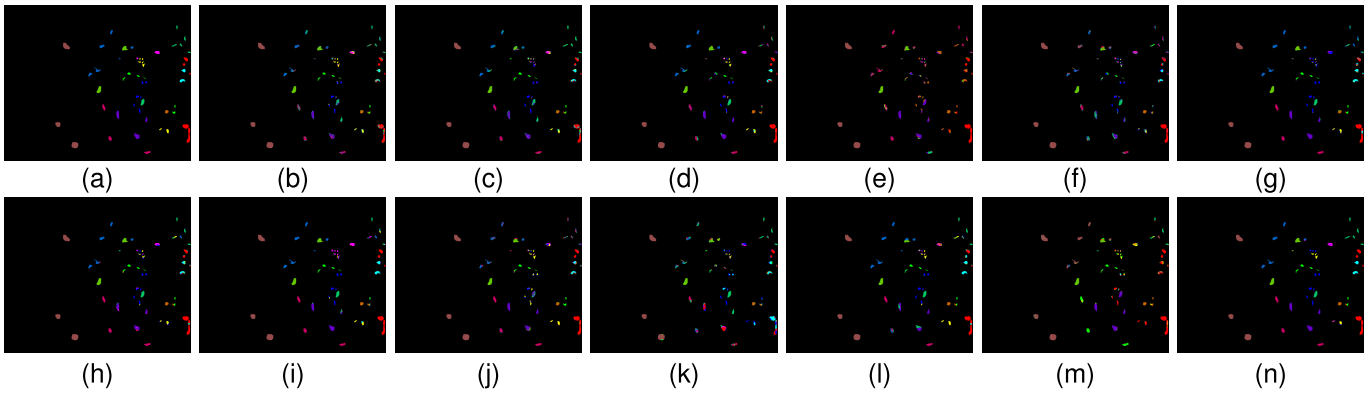


Fig. 4. Classification maps were generated for the KSC dataset, utilizing only ten labeled training samples per class. The corresponding representations are as follows. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) SSFTT. (l) CVSSN. (m) AMS. (n) Proposed.

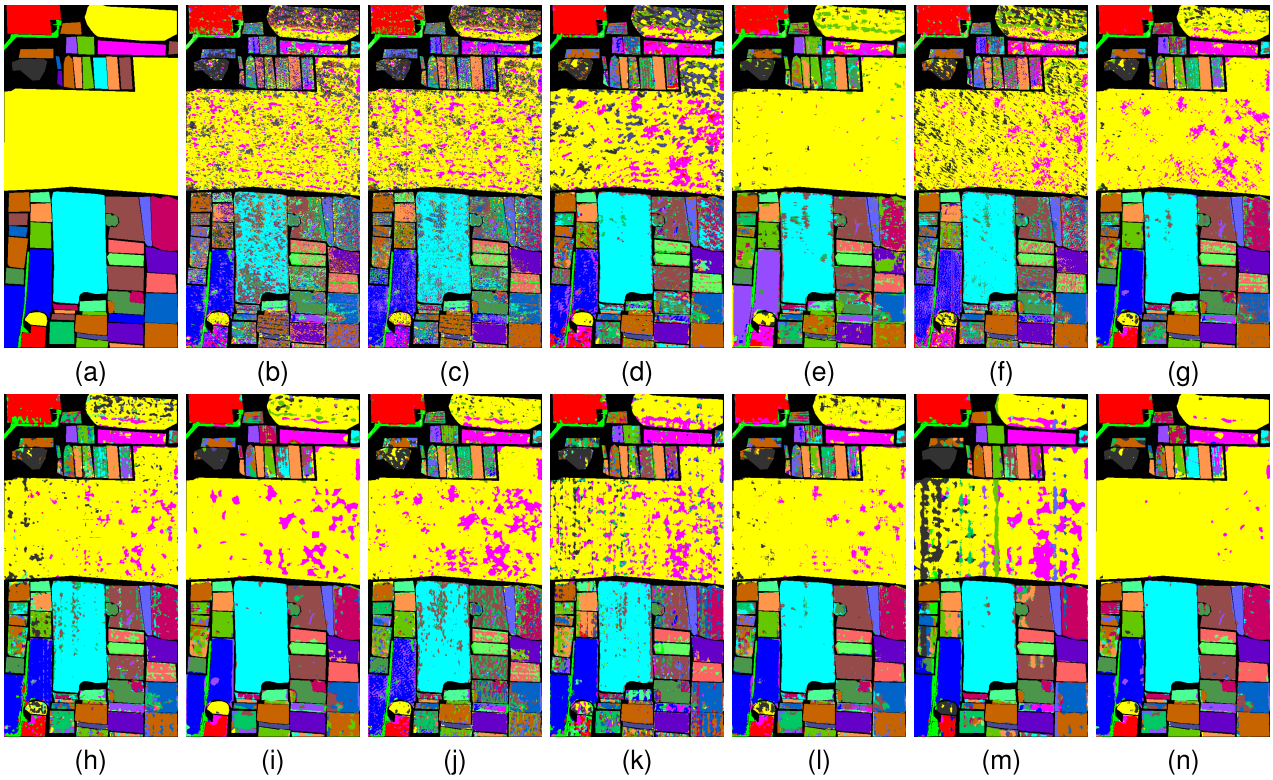


Fig. 5. Classification maps were generated for the Honghu dataset, utilizing only ten labeled training samples per class. The corresponding representations are as follows. (a) Ground-truth map. (b) RF. (c) SVM. (d) ContextNet. (e) SSAN. (f) RSSAN. (g) SSAtt. (h) A2S2K. (i) SSTN. (j) SSSAN. (k) SSFTT. (l) CVSSN. (m) AMS. (n) Proposed.

On the Honghu dataset, a resemblance to the classification patterns observed in the IP dataset is discernible across most algorithms. However, due to the Honghu dataset's larger central region and finer crop classification, a greater emphasis is placed on intermediate features, exacerbating spectral variations within the same feature. Consequently, certain deep learning algorithms exhibit misdivisions within this dataset. Yet, in all instances, our algorithm's classification outcomes are marked by smoother results and fewer instances of edge misclassification.

2) Efficiency Analysis: We conducted an efficiency analysis of different methods, and the results are presented in Table VII. The analysis encompasses factors, such as training/testing time, the number of model parameters, and computational cost. In terms of runtime, traditional methods, such as RF and SVM, exhibit relatively shorter durations for both model training and testing. Notably, the training process generally outpaces the testing phase due to its sensitivity to sample size. In contrast, deep learning-based algorithms display an inverse trend, characterized by longer training periods and shorter testing durations. Comparing the deep learning algorithms, our approach demonstrates noteworthy efficiency advantages. It stands out as faster during the training phase, especially on the Honghu dataset, and ranks only slightly behind the SSFTT algorithm on the IP and KSC datasets. Moreover, during the testing phase, our algorithm attains the shortest processing time among all deep learning-based methods.

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TABLE VII
COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP, KSC, AND HONGHU DATASETS

Dataset	metrics	RF	SVM	ContextNet	SSAN	RSSAN	SSAtt	A2S2K	SSTN	SSSAN	SSFTT	CVSSN	AMS	Proposed
IP	T_{train}	1.27±0.21	**0.04±0.03**	177.35±4.46	503.56±26.89	147.27±4.97	138.06±3.75	567.34±19.49	167.84±11.768	552.60±32.65	8.60±1.41	125.29±5.63	111.71±5.65	15.36±0.33
	T_{test}	**1.09±0.17**	1.58±0.24	7.24±1.15	11.97±1.52	10.88±0.77	10.53±1.09	4.54±0.10	11.58±1.64	106.85±4.11	1.59±0.04	6.30±0.71	65.86±0.71	**1.09±0.01**
	Params(M)	-	-	1.21	87.68	0.17	0.60	0.37	**0.02**	0.31	0.15	0.26	3.76	0.09
	FLOPs(M)	-	-	127.8	10548.91	11.76	22.50	256.71	**2.67**	26.74	6.99	31.46	4.15	10.66
KSC	T_{train}	0.82±0.06	**0.01±0.01**	132.18±8.46	334.22±21.55	137.19±4.77	138.06±2.65	470.86±7.19	161.34±6.77	78.92±13.22	8.40±0.32	91.66±11.63	81.38±1.31	9.17±0.68
	T_{test}	**0.38±0.13**	0.43±0.10	2.35±0.06	4.73±0.34	14.09±1.93	10.07±0.74	14.55±1.28	5.86±0.28	419.45±13.23	1.56±0.02	5.75±0.66	58.56±0.74	1.07±0.02
	Params(M)	-	-	1.07	68.75	0.15	0.58	0.33	**0.02**	0.30	0.15	0.26	2.95	0.09
	FLOPs(M)	-	-	114.46	8237.33	10.90	20.84	225.45	**2.52**	24.46	6.99	31.08	3.29	10.47
HH	T_{train}	1.84±0.30	**0.07±0.05**	79.68±3.345	783.70±16.90	61.84±2.63	79.02±6.21	843.37±19.75	94.04±7.56	271.18±26.20	11.28±1.01	68.90±9.76	230.49±24.05	9.7±0.33
	T_{test}	47.94±3.46	50.34±7.32	1.93±0.05	2.90±0.05	3.29±0.05	4.72±0.15	5.89±0.19	7.26±0.53	32.15±1.06	3.41±0.75	5.05±0.43	1.45±0.02	**1.44±0.03**
	Params(M)	-	-	1.59	156.36	0.22	0.64	0.48	**0.02**	0.34	0.14	0.27	6.79	0.07
	FLOPs(M)	-	-	166.26	18921.61	14.24	27.40	347.89	**3.08**	33.35	6.99	32.60	7.33	8.69

Regarding model size, we conducted a comprehensive assessment of parameter counts among deep learning algorithms. The provided table showcases both model size and computational workload for each algorithm. A clear observation emerges from the data: our proposed algorithm boasts one of the lowest parameter counts among all models examined. In terms of model parameters, our algorithm slightly surpasses only the SSTN algorithm. When considering computational volume, our algorithm remains notably efficient. Taken together with the outcomes of the earlier control experiments, we can confidently affirm that our algorithm attains exceptional classification performance while maintaining a comparatively modest model size and fewer parameters.

TABLE VIII
RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS

Dataset	Multi-res	C-att	ETF	OA	AA	Kappa
IP	✓	✓	✓	**85.40±1.57**	**76.68±2.69**	**83.46±1.75**
	✓	✓	-	83.08±1.76	75.74±3.15	80.86±1.95
	✓	-	✓	83.68±2.02	76.40±3.08	81.52±2.18
	-	✓	✓	73.81±2.89	68.71±2.65	70.63±3.03
KSC	✓	✓	✓	**93.44±1.86**	90.75±2.24	**92.68±2.07**
	✓	✓	-	92.98±5.67	**90.88±6.33**	92.22±6.22
	✓	-	✓	92.69±1.94	89.51±2.37	91.86±2.16
	-	✓	✓	89.76±2.80	86.65±6.81	88.57 ±3.13
HH	✓	✓	✓	**88.28±2.74**	**78.47±3.07**	**85.49±3.24**
	✓	✓	-	86.29±3.59	76.59±3.90	83.8±4.21
	✓	-	✓	86.02±3.23	76.83±2.81	82.83 ±3.74
	-	✓	✓	79.42±5.14	68.94±2.39	74.91±5.60

D. Ablation Study

To comprehensively demonstrate the effectiveness of our proposed module, we conducted ablation experiments across all three datasets. The primary objective was to retain the other modules while assessing the impact of removing each module on performance. The outcomes of these experiments are presented in Table VIII. The presence of the Multi-res flag indicates whether ResBlocks are embedded within the ResBlock module. When Multi-res is not enabled, we substitute the embedded ResBlock with a 2-D convolutional layer utilizing a 3×3 kernel. The C-att designation signifies whether channel attention is applied. Similarly, the ETF label indicates whether ETF classifiers are employed. In cases where the ETF classifier is not utilized, we replace it with a fully connected layer classifier, accompanied by the standard cross-entropy loss function. Within the table, a ✓ signifies the inclusion of a particular module. The ablation experiment results are presented in Table VIII, demonstrating that the removal of any module led to a degradation in classification performance across the three datasets. The optimal classification results were achieved when all three modules were preserved, further highlighting the efficacy of our proposed algorithm.

E. Parametric Analysis

The crucial parameters in our experiments mainly include patch size and the number of channels (filter number of CNN). The outcomes of varying filter numbers are depicted in Fig. 6. We conducted experiments with four different channel counts: 32, 64, 128, and 192. From the figure, it is evident that on the IP dataset, the optimal performance was achieved at 64 channels. As the channel numbers increased further, the classification accuracy did not exhibit a corresponding increase, and the AA decreased. For the KSC dataset, the classification accuracy reached its peak at 128 channels, and additional increments in the channel count led to a decline in all three quantitative indicators of classification accuracy. On the Honghu dataset, favorable OA and kappa values were attained at 64 and 192 channels, with AA showing an increase as the channel count increased. It is important to note that an increase in the number of channels would result in elevated computational load and parameter count. Consequently, for our primary experiments, we opted for 64 channels in the network architecture.

Regarding the patch size, we conducted an analysis ranging from 7×7 to 15×15 on all three datasets. The results are presented in Fig. 7. Notably, a patch size of 11×11 yielded favorable outcomes for both the IP dataset and the Honghu dataset. However, on the KSC dataset, the patch size of 13×13 demonstrated improvements compared with 11×11. It is worth highlighting that larger patch sizes impose higher demands on memory and computational resources. In the case of the KSC dataset, which exhibits less noise and more scattered category distribution, opting for larger blocks typically avoids cross-category spatial intersections. Consequently, larger blocks provide enhanced information and yield better results. In contrast, the IP dataset contains more noise, and given the proximity of the IP and Honghu categories, larger blocks are more susceptible to containing noise or capturing edge areas. As a result, larger blocks lead to diminished

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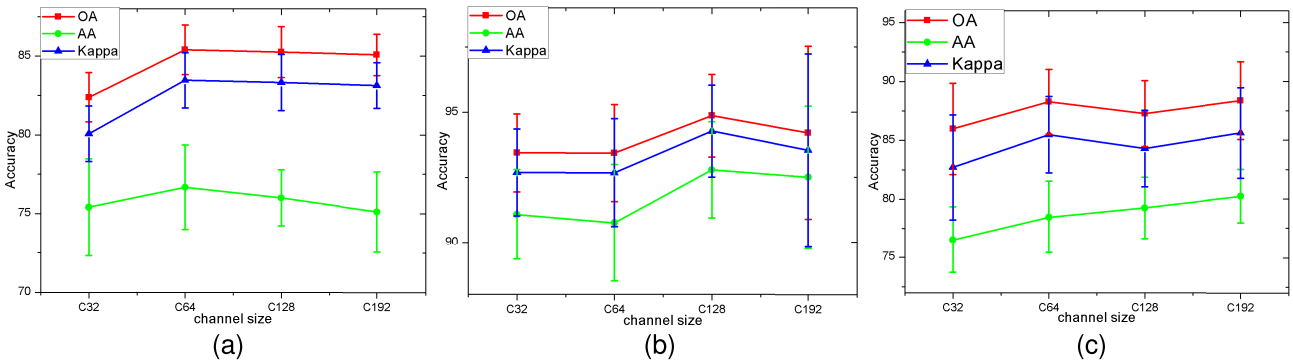


Fig. 6. Classification accuracy of different filter channels over the three datasets (a) IP. (b) KSC. (c) Honghu.

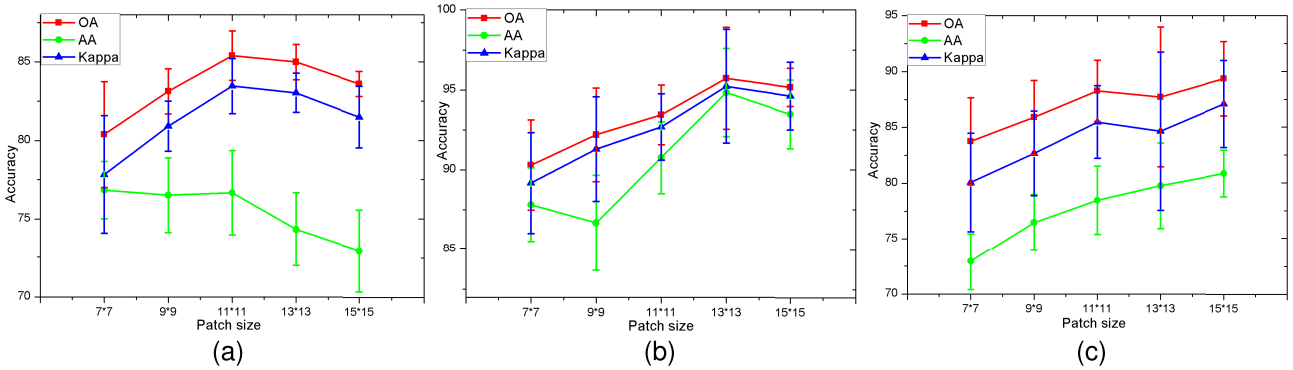


Fig. 7. Classification accuracy of different patch sizes over the three datasets.(a) IP. (b) KSC. (c) Honghu.

TABLE IX
RESULTS OF DIFFERENT NUMBERS OF SAMPLES

Metrics	1	5	10	20	50	100
OA	59.56±9.45	82.83±3.92	88.28±2.74	91.75±0.99	94.60±0.96	96.17±3.09
AA	50.30±5.68	71.21±3.76	78.47±3.07	83.14±1.70	88.76±1.25	92.43±1.13
Kappa	52.36±10.32	78.97±4.30	85.49±3.24	89.70±1.21	93.24±1.17	95.25±3.66

performance. For consistency across the three datasets in our primary experiments, we employed uniform 11 × 11 block sizes.

In addition, we also present the classification results of HSIs under different sample conditions. Since the number of samples in the IP and KSC datasets is limited, we only show the results on the HH dataset. In the experiments, we varied only the number of training samples while keeping the number of testing samples constant. The obtained results are shown in Table IX. As seen from the table, when the number of samples per class is only 1, our proposed algorithm still achieves a classification accuracy of 59.56%. With an increase in the number of samples, the classification accuracy improves. When using 200 samples, the accuracy increase is not substantial, mainly because the accuracy is already quite high. Moreover, for different numbers of samples, there should be different optimal hyperparameters. Here, we use this solely for illustrating that our algorithm performs well under various sample conditions.

V. CONCLUSION AND DISCUSSION

The challenge of limited samples for hyperspectral classification often arises when acquiring a significant number of labeled hyperspectral samples is difficult. In response, we have introduced a novel hyperspectral classification network. Key components include ResBlock-in-ResBlock, which enhances nonlinear representation, and the application of channel attention to learned residual features, thereby further refining feature representation based on preceding modules. Drawing inspiration from the concept of neural collapse, we have employed a fixed classifier with point loss for the final classification. Our experimental results across various HSI datasets demonstrate that our algorithm enhances classification accuracy. The ablation experiment verifies the effectiveness of each proposed module.

After analyzing complexity and parameter count, it becomes evident that our algorithm boasts fewer actual parameters and computational complexity in contrast to the majority of other deep learning algorithms. While this article employs a straightforward fusion of a basic backbone feature extraction network and an ETF classifier, architectural exploration could yield more sophisticated backbone networks tailored to HSI classification. Furthermore, despite the scarcity of labeled samples, a wealth of unlabeled samples exists. Harnessing these unlabeled samples presents a promising avenue for tackling forthcoming challenges in the realm of small sample classification.

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